

RESEARCH ARTICLE

Temperature dependent charge transport in electrostatically doped poly[benzimidazobenzophenanthroline] thin films

Alejandro J. Cruz-Arzon¹ | William Serrano-Garcia² | Nicholas J. Pinto¹  | Nikita Gupta³ | Alan T. Charlie Johnson³

¹Department of Physics and Electronics, University of Puerto Rico at Humacao, Humacao, Puerto Rico, USA

²Department of Electrical Engineering, University of South Florida, Tampa, Florida, USA

³Department of Physics and Astronomy, University of Pennsylvania, Philadelphia, Pennsylvania, USA

Correspondence

Nicholas J. Pinto, Department of Physics and Electronics, University of Puerto Rico at Humacao, Humacao, PR 00791, USA.
Email: nicholas.pinto@upr.edu

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Abstract

Charge transport in electrostatically doped poly[benzimidazobenzophenanthroline]-BBL thin films in a field-effect transistor geometry were investigated in the temperature range $150\text{ K} < T < 370\text{ K}$. At low temperatures activation and hopping transport mechanisms dominated, while phonon scattering dominated at high temperatures. The activation energies (E_A) were found to lie in the range $140\text{ meV} < E_A < 400\text{ meV}$ implying the existence of deep traps within the polymer bandgap of 1.8 eV . Two quasi-linear dependencies of E_A on the gate voltage (V_g) were observed with E_A decreasing as V_g increased. An unexpected “metallic-like” transport characteristic appeared for $T > 335\text{ K}$ which depended on V_g . Enhanced electron delocalization combined with increased carrier density could be responsible for this “metallic-like” behavior. Our results show that the existence of deep traps with multiple energy distributions, combined with increased carrier density led to the unusual temperature dependence of charge transport observed in BBL.

KEYWORDS

conducting polymers, crystallization, kinetics, polyimides, thermal properties

1 | INTRODUCTION

Organic conducting polymers play an important role in plastic electronics.^{1–4} Early studies led to the synthesis of organic polymers that were electrically conducting but proved to be insoluble and unstable in air.^{5,6} New routes to chemical synthesis and post processing however have now resulted in these polymers being soluble in common organic solvents in the conducting form.^{7–10} This avoids the use of cumbersome methods like thermal evaporation to prepare thin films. In addition, many commercially available conducting and semiconducting polymers (both *p* and *n*-type) are chemically stable under normal laboratory conditions. These features

helped expand the use of conducting polymers to include new applications ranging from passive rust protection in metals,^{11–14} in the field of medicine,^{15,16} to electrically active applications such as in diodes, sensors, photovoltaics, and field effect transistors.^{17–22}

Improving the electronic properties of these polymers involves understanding the mechanisms that govern charge transport. Many studies on the temperature dependent charge transport have been published on single strand, linear chain conducting and semiconducting polymers.²³ Charge hopping between localized states, small polaron formation in conjugated systems, multiple trap and release and phonon scattering are some of the transport mechanisms invoked to explain the

temperature dependence of the electrical conductivity.^{23,24} Very few temperature-dependent charge transport studies exist however on double-stranded or ladder type conducting polymers.^{25,26} Of the few reports available, the focus was on the undoped polymers. These polymers usually have a discrete layered structure, where in thin films the layers align nearly perpendicular to the substrate^{27,28} resulting in considerable anisotropy in the conductivity (i.e., conductivity along the film versus conductivity perpendicular to the film).^{25,26} They are more crystalline and planar due to the double strands that hinder backbone twisting, and which in turn promotes greater charge delocalization.²⁹ In addition, the presence of alternating bond lengths and of multiple conjugation paths that are available to bypass defects differentiate them from single stranded conducting polymers. Hence charge transport in these polymers could display characteristics not typically found in their single stranded counterparts. Also, the nature of the electronic excitations is still poorly understood in these systems.³⁰ A need therefore exists to investigate charge transport in ladder type conducting polymers.

Herein we report on electrical characterization of the *n*-type conjugated ladder polymer poly[benzimidazobenzophenanthroline]-BBL. BBL was chosen specifically for this study because of its high thermal stability in air³¹ and due to an absence of temperature dependent charge transport studies on this polymer in the electrostatically doped state. We used a thin film field-effect transistor (FET) device architecture and conducted experiments under different electrical bias and temperature conditions in the range $150\text{ K} < T < 370\text{ K}$. Activation, hopping, and phonon scattering were three mechanisms that governed charge transport. The device output characteristics showed clear current (I_{ds}) saturation at high and at low temperature, and the device transconductance characteristics showed an increasing I_{ds} with increasing positive gate voltage (V_{g}), consistent with the *n*-type nature of the polymer.¹⁰ The temperature dependence of the current also varied with V_{g} . For high V_{g} ($>20\text{ V}$), $I_{\text{ds}}(T)$ was non-monotonic, increasing with increasing temperature up to T_{max} above which it decreased. A decrease in I_{ds} for $T > T_{\text{max}}$ resembled “metallic-like” diffusive transport, and the temperature at which I_{ds} was a maximum shifted to higher values with lowering V_{g} . One of the traditional signatures of metallic transport in conducting/semiconducting polymers is the negative temperature coefficient of the conductivity (σ) that is, $\frac{d\sigma}{dT} < 0$.³² We used the term “metallic-like” to describe charge transport in BBL in that range of temperature where the current I_{ds} flowing through BBL decreased as the temperature increased. Inherently planar BBL with enhanced electron delocalization combined with increased carrier density

likely facilitated the “metallic-like” behavior. The temperature dependence of the charge mobility (μ) was non-monotonic as well indicating the presence of two charge transporting regimes. Further analysis of the low temperature dependence of the channel conductance (G) revealed two quasi-linear dependencies of the activation energy on V_{g} . This suggested the presence of multiple energy distributions of trap states within the band gap of the polymer. Our results show that the existence of deep traps with multiple energy distributions, combined with increased carrier density and enhanced charge delocalization led to the unusual temperature dependence of charge transport observed in BBL.

A few conducting polymers have shown such behavior. Polyaniline and TIPS pentacene are two *p*-type conducting polymers that have demonstrated “metallic-like” behavior that is, the conductivity or mobility exhibited a negative temperature gradient. In the case of polyaniline, two models were proposed to explain the origin of the “metallic” behavior. In the first it was assumed that the doping process (i.e., protonation) was random which resulted in a homogeneous distribution of protonation sites along the polymer chains.^{23,32} Thus at high protonation levels, conduction band states would overlap and give rise to “metallic” behavior. The second model assumed a non-homogenous doping process which resulted in highly conducting “metallic” islands randomly distributed in an amorphous poorly conducting matrix.³³ In this case, for highly doped polyaniline, the number and size of these “metallic” islands increased resulting in increased “metallic” material, shorter inter-metallic distances, and increased charge carrier density. The combined effect gave rise to the observed “metallic-like” behavior. In the case of TIPS pentacene, the “metallic-like” behavior was not attributed to extended state conduction but due to localized transport that was limited by phonon scattering.³⁴ A “metallic-like” behavior in electrostatically doped *n*-type conducting polymers has not been reported, which makes the present study on BBL unique.

2 | EXPERIMENTAL

BBL and methane sulfonic acid (MSA) were purchased from Millipore Sigma and used as received. BBL was a brown powdered material having a solubility of 1 ml/mg in MSA. Depending on the synthesis conditions, BBL with molecular weights in the range 4.9 to 51 kDa have been synthesized.³⁵ In this work, a 2.0 mg/ml BBL in MSA was prepared which resulted in a uniform solution that was red in color to the naked eye. Due to the non-wettability of MSA on the Si^+/SiO_2 substrate and its high

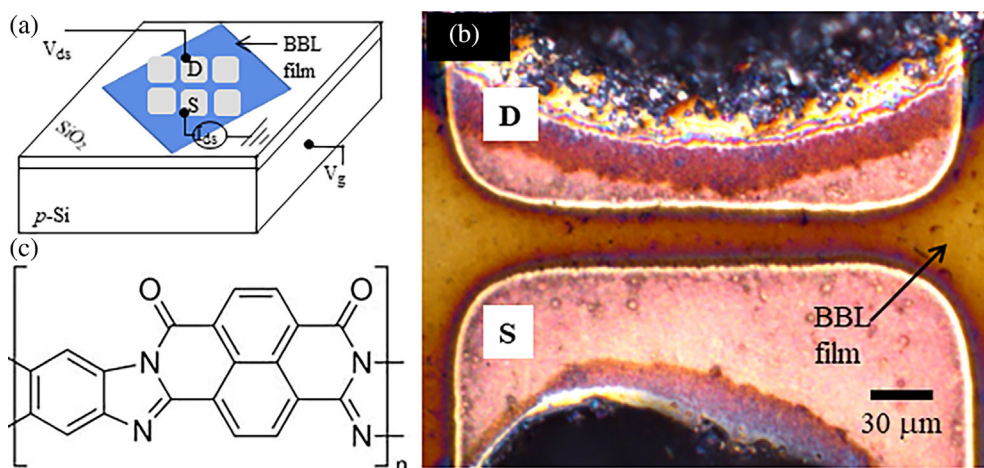


FIGURE 1 (a) Schematic of the device showing the Si^+/SiO_2 substrate, the main components and the external connections used to electrically characterize it. The gray metal squares rounded at the ends, S and D, represents the thermally evaporated silver source and drain contacts of the device respectively. (b) A magnified top view optical microscope image of the S/D terminals (seen in (a)) of the actual device. The black material partially covering S and D is silver epoxy that was used to hard wire the device. (c) Chemical structure of the repeat unit in double stranded BBL- $(\text{C}_{20}\text{H}_6\text{N}_4\text{O}_2)_n$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53470)]

boiling point, we found it difficult to spin coat thin films under ambient conditions and hence used a different approach. Two pieces of Si^+/SiO_2 (300 nm oxide) wafers ($1'' \times 1''$) were cut and a drop of BBL/MSA placed on one piece. The other piece was then carefully placed over it sandwiching the drop. A normally closed tweezer was used to hold the two pieces together and then submerged vertically in a beaker of deionized water for 72 h. Gradual diffusion of water and removal of MSA assisted by gravity resulted in thin films slowly solidifying between the two substrates, and the color of the films changed from red to blue. The film crystallinity using this method was expected to be higher than those formed via spin-coating techniques where rapid evaporation of the solvent occurs. Due to slow method of film formation, we assumed an “edge-on” conformation of polymer chains on the substrate such that the π - π overlap was parallel to the substrate.²⁸ The substrates were then placed in an oven at 165°C for 1 hour to remove any MSA traces and to ensure that the films firmly adhered to the substrate. After separating the substrates, thin films of BBL with varying thicknesses were seen to be randomly distributed over both surfaces. Uniform films of area approx. 1×1 mm were used for device fabrication.

To fabricate the drain (D) and source (S) electrodes of the device, a transmission electron microscope (TEM) grid was used as a shadow mask and placed over a preselected uniform thin film as fabricated above. The doped Si^+ substrate served as the gate (G) electrode. Using a thermal evaporator, 100 nm of Ag was evaporated through the mask. The region between the S/D terminals represented the BBL channel. Figure 1a,b shows a

schematic of the FET device with electrical connections, and a top view microscope image of the actual device used in this work respectively. Figure 1c shows the chemical structure of BBL. The channel length (L) and width (W) were 0.03 mm and 0.25 mm respectively. The average film thickness was measured to be 5 μm . The device was hard wired with Au wires and Ag epoxy, and electrical measurements were made in vacuum using a Keithley Model 6517B electrometer for V_{ds} and I_{ds} , and a Keithley Model 2400 source meter for V_{g} . A CTI-Cryogenics compressor was used to vary the temperature that was stabilized by a CryoCon 32B temperature controller. The temperature was allowed to stabilize to ± 0.2 K prior to performing an $I_{\text{ds}}-V_{\text{g}}$ scan. For the $I_{\text{ds}}-T$ measurements, the temperature was scanned at a rate of 2 K/min. A Mitutoyo digital micrometer was used to measure the film thickness.

3 | RESULTS AND DISCUSSION

Figure 2 shows the device output characteristics at three different temperatures. The increase in I_{ds} with increasing V_{g} and its subsequent saturation as V_{ds} increased, demonstrated the n -type nature of BBL.¹⁰ Good current saturation was observed at high V_{ds} even at low temperatures implying FET accumulation mode of operation.¹⁰ Lowering the temperature was also seen to require a larger V_{g} for channel conduction. The inset to the graph at 295 K represents I_{ds} versus V_{ds} at zero gate bias plotted on an expanded current scale. This plot confirmed the electronic (Ohmic) nature of the measured conductivity

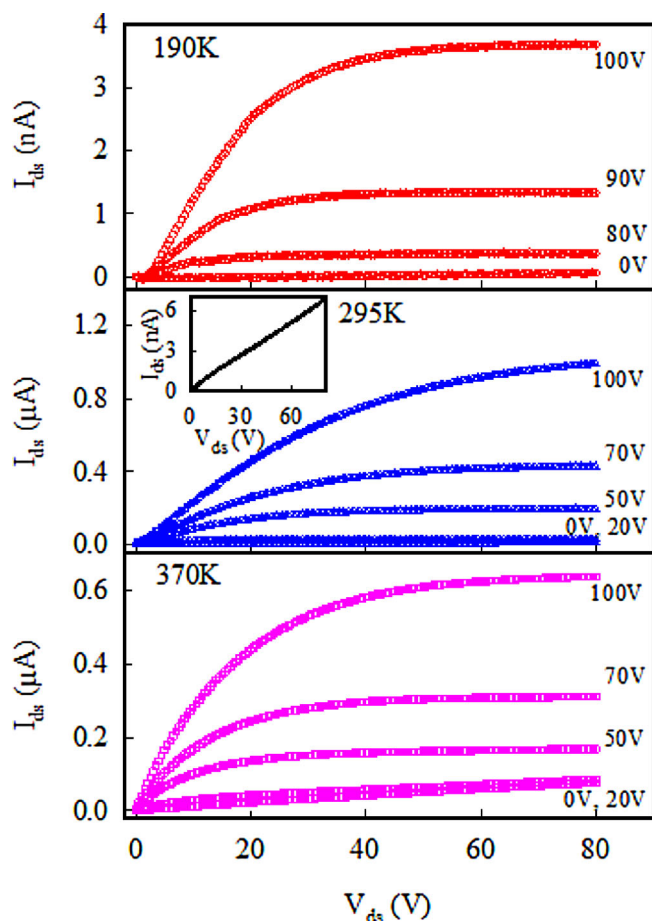


FIGURE 2 Drain-source current (I_{ds}) versus drain-source voltage (V_{ds}) for selected gate voltages (V_g) shown to the right of the curves, measured at 190, 295, and 370 K. The inset to the middle graph shows I_{ds} versus V_{ds} with $V_g = 0$ V at 295 K on an expanded current scale. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

and that any Schottky barriers formed at the metal electrode/polymer interface were negligible. From the I_{ds} – V_{ds} traces in the absence of charge injection/electrostatic doping ($V_g = 0$) the conductivity along the film ($\sigma_{//} = \frac{L}{AR}$) was calculated to be 2.6×10^{-10} , 1.8×10^{-8} , and 2.6×10^{-7} S/cm at 190, 295, and 370 K respectively, where ($R = \frac{V_{ds}}{I_{ds}}$) was the channel resistance, and A the channel cross-sectional area. Similar values were measured for the conductivity along the BBL film.²⁵ The instant rise in I_{ds} at small V_{ds} as seen in Figure 2 for $T = 370$ and 295 K was independent of V_g and implied that the effects of contact resistance were negligible. A slightly delayed response at 190 K could mean that the contact resistance at low temperatures and low V_{ds} began to affect charge transport by limiting charge injection into the channel. We do not believe that the contact resistance had a significant impact on the charge transport mechanisms investigated in this report. Figure 2 also

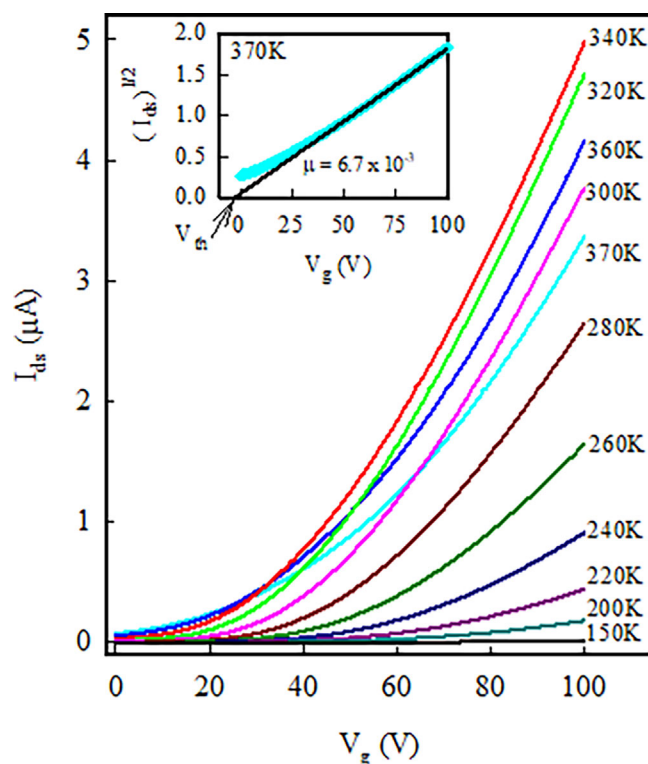


FIGURE 3 Drain-source current (I_{ds}) versus gate voltage (V_g) for fixed $V_{ds} = 80$ V plotted for a few selected temperatures shown to the right of each curve. Inset: $\sqrt{I_{ds}}$ versus V_g at 370 K. The slope of the straight-line fit to the linear section of this plot was used to find the mobility ($\mu = 6.7 \times 10^{-3}$ cm²/V-s) and the voltage intercept at zero current yielded the threshold voltage ($V_{th} = -2$ V) as per Equation (1). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

shows that the saturated currents at 295 K were slightly higher than those measured at 370 K under similar bias conditions, a feature that will be discussed later.

Figure 3 shows the device transconductance characteristics (I_{ds} – V_g), for selected temperatures in the range $150 \text{ K} < T < 370 \text{ K}$. BBL is known to be stable up to 600°C in air and up to 700°C in an N_2 environment,³¹ hence we do not expect the BBL film to undergo thermal degradation in our experiment. The device was heated to 370 K, and after current and temperature stabilization, each I_{ds} – V_g measurement was made at 5 K temperature intervals while cooling. V_{ds} was fixed at 80 V to ensure that the device operated in the accumulation and saturation mode during the measurements, as observed in Figure 2. Figure S1 in the Supplementary section shows I_{ds} – V_g plots at the other recorded temperatures in a 3-D format. An initial increase in I_{ds} was observed as the temperature was lowered from 370 K, consistent with the I_{ds} – V_{ds} plots in Figure 2. In the saturated regime of device operation, the channel current is given by³⁶:

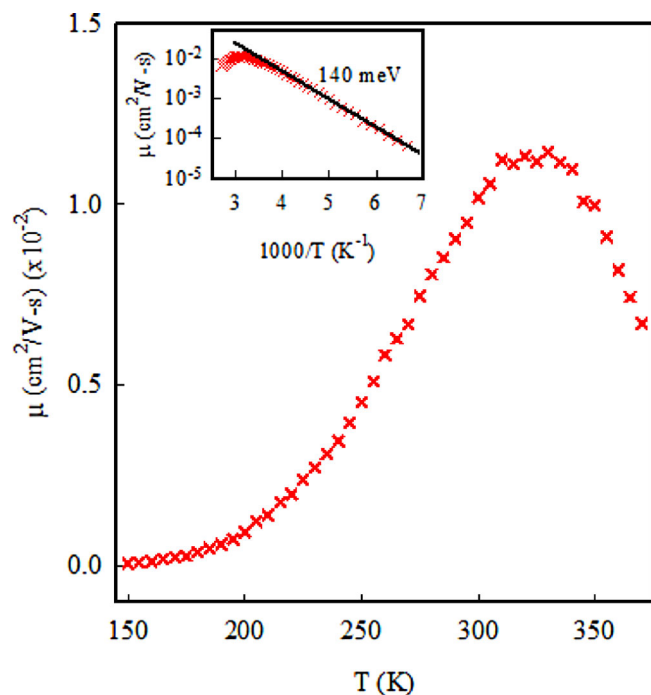


FIGURE 4 Temperature dependence of the mobility (μ). Inset: Arrhenius plot of the mobility as a function of inverse temperature. The activation energy of 140 meV was calculated using Equation (2). [Color figure can be viewed at wileyonlinelibrary.com]

$$I_{ds} = \frac{\mu W C_i}{2L} (V_g - V_{th})^2 \quad (1)$$

Where μ is the charge(electron) mobility at high carrier charge concentrations, C_i is the oxide geometrical capacitance (11.5 nF/cm^2), and V_{th} is the threshold voltage representing the onset of rapidly increasing conduction in the channel. The inset to Figure 3 shows $\sqrt{I_{ds}}$ versus V_g using the data taken at 370 K as a representative plot. The slope of the best fit line to the linear section in this plot yielded μ while the voltage intercept at zero current yielded V_{th} as per Equation (1). Figure S2 shows that the linear behavior in the $\sqrt{I_{ds}}$ versus V_g plots for $V_g > V_{th}$ was well defined down to 150 K, ensuring the reliability of the computed values of μ and V_{th} .

As in single stranded conducting polymers, the likely mechanisms of charge transport in BBL include charge hopping between localized states, multiple trap and release of charges via thermal activation, and phonon scattering.^{23,24} The first two mechanisms increase the current with increasing temperature while phonon scattering decreases the current with increasing temperature. The first two mechanisms typically dominate at low temperatures while phonon scattering dominates at high temperatures. Since BBL is double stranded, the fundamental repeat unit has a larger surface area, varying bond

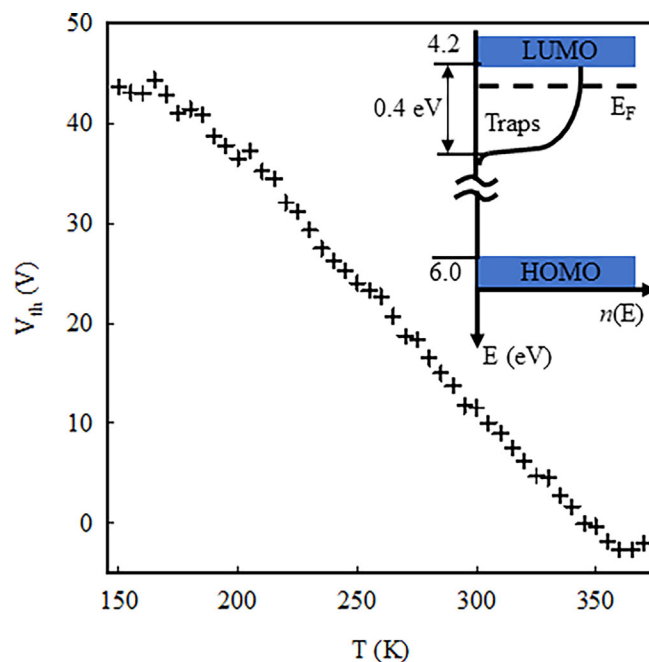


FIGURE 5 Threshold voltage (V_{th}) as a function of temperature. Inset: Suggested energy band diagram of the electronic states near the LUMO level in the BBL thin film FET based on values of the activation energy calculated in this report. The energy values of the LUMO and HOMO levels shown in the figure were obtained from Ref. 30. [Color figure can be viewed at wileyonlinelibrary.com]

lengths and multiple conjugation paths.²⁵ Defects can hence be bypassed, furthermore the increased planarity of BBL due to the double strands means greater charge delocalization. BBL also has a discrete layered structure due to efficient π - π stacking, and the polymer backbone is arranged such that the π - π overlap is parallel to the substrate^{27,37} enhancing the in-plane conductivity. Effects of substrate phonons on charge transport are therefore expected to be pronounced in BBL thin films. These features of double stranded BBL suggest that charge transport as a function of temperature may reveal characteristics not seen in single stranded conducting polymers.

Figure 4 shows the temperature dependence of μ , and its maximum value of $1.1 \times 10^{-2} \text{ cm}^2/\text{V-s}$ was comparable to previous reports on BBL.^{10,29} The mobility was seen to stay constant at this value in the temperature range $310 \text{ K} < T < 335 \text{ K}$. This temperature range of constant μ was not unique as another device investigated showed that it occurred in a different temperature range. Its low value was likely due to disorder at the polymer/substrate interface and/or the presence of charge traps. Applying a low-k dielectric buffer layer to SiO_2 prior to depositing the BBL film can improve mobility.³⁸ The non-monotonicity in $\mu(T)$ suggests the presence of two

charge transport regimes.³⁹ Above 335 K, $\mu(T)$ decreases with increasing T which we attributed to phonon scattering, while below 310 K, $\mu(T)$ decreases with decreasing T and was primarily due to charge hopping and charge activation via trap and release. For data analysis, the mobility was fitted with an Arrhenius dependence³⁹:

$$\mu(T) = \mu_0 \exp\left(-\frac{E_A}{k_B T}\right) \quad (2)$$

Where μ_0 is the pre-exponential factor and E_A the activation energy. The upper inset to Figure 4 shows an Arrhenius plot of μ versus $\frac{1}{T}$ from which the activation energy (for $V_g > V_{th}$) was calculated to be 140 meV using Equation (2). Below 310 K, we argue that charge transport was mainly dominated by multiple trap and release of carriers by trap states located within the HOMO-LUMO band gap of the polymer and by charge hopping between traps. The traps are localized electronic states that are associated with polymer defects (chemical impurities/structural disorder), electronic stress, and surface states. Most of these states typically lie within a few 100 meV below the LUMO level since the measured mobility is rather low (deep traps). The activation energy calculated above represented the energy required to excite the gate induced electrons in the channel that are localized in the traps into the LUMO band.

Figure 5 shows the temperature dependence of V_{th} which increased quasi-linearly as temperature was lowered and was consistent with reports on other conducting polymers.⁴⁰ Lowering the temperature made it harder for the device to turn on and could be understood in terms of charge trapping. Higher temperature implied higher energy ($k_B T$) that could overcome low lying energy traps. Upon cooling, this energy was lowered, and a higher V_g was necessary to increase the injected charge concentration and fill the deep traps. This in turn led to a higher threshold voltage. The concentration of trap states ($N_{tr} = \frac{C_i V_{th}}{e}$) with energies $> k_B T$ below the LUMO level increased from $1 \times 10^{11} \text{ cm}^{-2}$ at 370 K to $3 \times 10^{12} \text{ cm}^{-2}$ at 150 K. The quasi-linear response of $V_{th}(T)$ with lowering temperature as seen in Figure 5 suggested that the energy distribution of trap states $\left[\frac{\partial N_{tr}}{\partial E} = \left(\frac{C_i}{k_B e}\right) \left(\frac{\partial V_{th}}{\partial T}\right) = 2.1 \times 10^{14} \text{ cm}^{-2} \text{ eV}^{-1}\right]$ was energy independent within a fraction of $k_B T$ below LUMO level,³⁹ while above $k_B T$ there was an energy distribution of traps. The LUMO-HOMO band gap for BBL was reported to be 1.8 eV³⁰ and the inset to Figure 5 shows a likely energy-band diagram of the electronic states near the LUMO level.

To gain additional insight into the dynamics of charge transport, the temperature dependence of the channel conductance ($G = \frac{1}{R}$) was analyzed. Using the data in Figure 3 and Figure S1, and assuming a temperature

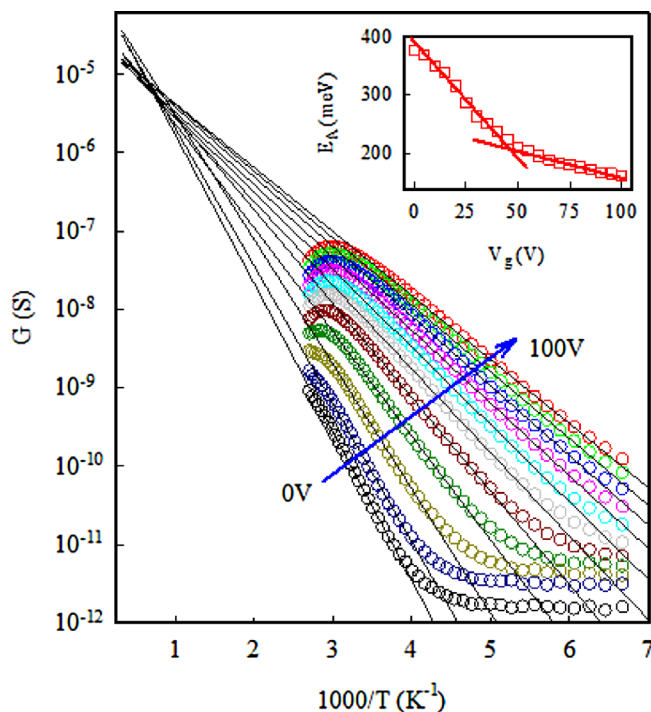


FIGURE 6 Arrhenius plots of the channel conductance (G) as a function of inverse temperature (T) for various gate voltages: 0 V $\rightarrow$ 100 V in 10 V steps. The straight black lines are fits to the linear portion of the curves after channel turn-on from which E_A was calculated using Equation (2) with μ is replaced by G . Inset: Activation energy (E_A) plotted as a function of gate voltage (V_g). Two quasi-linear sections are seen as indicated by the two red lines. [Color figure can be viewed at wileyonlinelibrary.com]

dependence for G that was equivalent to Equation (2), Arrhenius plots of G as a function of inverse temperature for various V_g ranging from 0 to 100 V are shown in Figure 6. From the slopes of the straight-line fits to the linear section of the data, the activation energies (E_A) were calculated and plotted as a function of V_g in the upper inset to this Figure. The activation energies were found to lie in the range $100 \text{ meV} < E_A < 400 \text{ meV}$ implying the presence of deep traps. Shallow traps typically require activation energies $< 100 \text{ meV}$ for excitation into the conduction or valence band.⁴¹ Since the carrier charge injected (with density $n = \frac{C_i(V_g - V_{th})}{e}$) into the channel depended on V_g , the different slopes in Figure 6 indicated that the activation energy also depended on n and got smaller with increasing n . Thus the activation energy was seen to decrease monotonously with increasing V_g .

A closer look at the inset to Figure 6 shows two quasi-linear regions in the dependence of E_A on V_g . The rate at which E_A decreases with increasing V_g is an indicator of the density of trap states near the Fermi level.⁴² When

the straight-line fits seen in Figure 6 were extrapolated toward higher temperatures, no intersection at a single point was observed. The absence of a well-defined isokinetic point that would yield a characteristic temperature suggested the presence of multiple charge transport mechanisms.⁴¹ Figure S3 shows plots of G versus $1/T$ (activation) and of G versus $\frac{1}{\sqrt{T}}$ (quasi-1-D hopping). Both showed linear dependencies, but it was not possible to say with certainty which process governed charge transport. A discrete layered crystal structure, distribution of bond lengths and various conjugations paths in BBL could be some reasons for the presence of multiple charge transport mechanisms. We argue that multiple exponential distributions of energy trap states exist in BBL. The gate voltage controlled the way these traps were filled, and together with competing transport mechanisms gave rise to the quasi-linear features observed in the E_A (V_g) plot. In the inset to Figure 5, the energy band diagram shows an energy independent trap state density near the LUMO level. Above this, the energy distribution of trap states should exhibit multiple exponential dependencies. Below V_{th} , channel conduction was negligible and was due to residual charges present in the polymer. At low V_g ($>V_{th}$) n is small, and the injected carrier charges were localized in deep traps. Phonon assisted charge hopping and a high activation energy associated with the trap and release mechanism combined to excite charges into the LUMO band. With increasing V_g however, charge (electron) injection at the S terminal of the device increases, that is, n is large, and the traps get filled. The Fermi level moved through the trap states getting closer to the LUMO band edge.³⁹ Also, fewer trap states existed near the Fermi level for large V_g . Hence the channel conductivity increased rapidly due to thermal excitation of charges into the LUMO band. The activation energy for high V_g was therefore smaller. This explains the two quasi-linear regions seen in the inset to Figure 6. We note that the Arrhenius plot of the mobility lacked the two quasi-linear sections seen in the inset to Figure 6. This was because the mobility was independent of V_g and was calculated when n was high ($V_g > V_{th}$). The activation energy of 140 meV calculated using $\mu(T)$ for high n was consistent with the low values of E_A calculated using $G(T)$ (see inset to Figure 6).

We next analyzed charge transport at high temperatures. Figure 3 showed an initial increase in I_{ds} as the temperature was lowered from 370 K. As a result, in Figure 4 the mobility was seen to decrease as temperature increased beyond 335 K. The upper inset to Figure 4 shows a high temperature roll-off from Arrhenius behavior. Such an effect has been observed in other n -type polymers and was attributed to the high temperature thermal energies being comparable to the trap depth.⁴¹

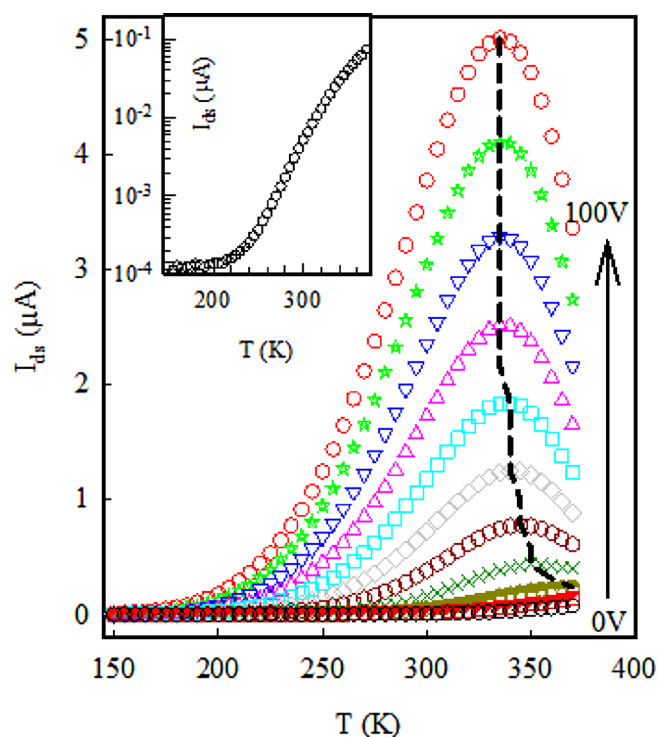


FIGURE 7 Drain-source current (I_{ds}) as a function of temperature (T) for different values of V_g ranging from 0 to 100 V in 10 V steps as shown in the direction of the arrow. These data points were extracted from Figure 3 and Figure S1 where V_{ds} was held fixed at 80 V during the measurements. The dashed line is a guide to the eye showing the dependence of the peak current versus T for each gate voltage. Inset: I_{ds} versus T for $V_g = 0$ V plotted on a semi-log scale. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

The presence of a negative temperature coefficient of the mobility ($\frac{d\mu}{dT} < 0$) suggests “metallic-like” transport.³⁴ The maximum conductivity measured at $V_g = 100$ V was 10^{-5} S/cm (see Figure 6), less than the minimum metallic conductivity criterion [100–300 S/cm].^{43,44} Also, from Figure 2, the BBL channel exhibited n -type semiconductor characteristics even at high temperatures. Hence the “metallic-like” behavior seen above 335 K was not anticipated. Double stranded BBL was inherently planar with multiple intralayer conjugation paths, and the resulting enhanced electron delocalization combined with increased carrier density could facilitate “metallic-like” behavior. However, rather than observing band transport, perhaps raising the temperature increased movement of the polymer backbone, resulting in greater interchain phonon backscattering of charge carriers which reduced the current (i.e., localized charge transport limited by phonon scattering). The precise origin of this effect remains unclear.

Since the mobility was derived from I_{ds} (V_g), we replot the data shown in Figure 3 and Figure S1 and study the

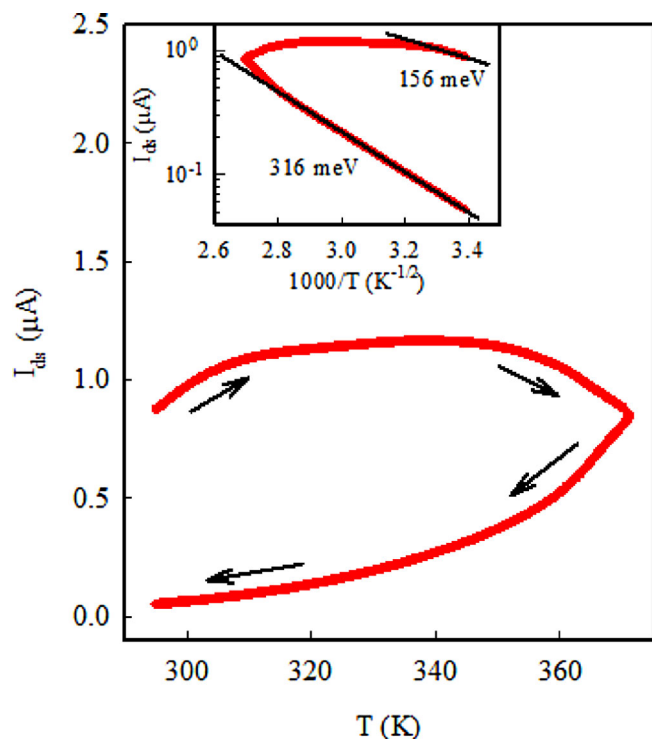


FIGURE 8 Temperature dependence of I_{ds} in the temperature range $295\text{ K} < T < 370\text{ K}$ during heating and cooling run as indicated by the arrows. The device bias condition for this I_{ds} versus T measurement was: $V_{ds} = 80\text{ V}$ and $V_g = 100\text{ V}$. Inset: Arrhenius plot of the data shown in the main figure. The activation energies were calculated using Equation (2) in the manuscript with μ replaced by I_{ds} . [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

temperature dependence of I_{ds} to better understand the “metallic” behavior. Figure 7 shows the temperature dependence of I_{ds} for various gate voltages V_g with $V_{ds} = 80\text{ V}$. The curves show a peak value in the observed current for several V_g . The temperature (T_{max}) at the peak values of the current stayed relatively constant at 335 K for $100\text{ V} < V_g < 50\text{ V}$, and the peak then shifted to higher temperatures with smaller V_g as seen by the dashed line. For $V_g > 50\text{ V}$, n was high which resulted in traps being filled. Trap filling brought the Fermi energy level closer to the LUMO band edge as mentioned earlier. The activation energy needed to excite these charges into the LUMO band was weakly dependent on temperature since the thermal energy was close to that of the trap depth and the energy distribution of traps was relatively constant near the LUMO level. Hence the peak temperature stayed relatively constant for high V_g , and phonon scattering dominated charge transport leading to decreased current with increasing temperature. For $V_g < 50\text{ V}$, n decreased, and charges were localized in deep traps requiring a higher activation energy for conduction. The thermal energy needed for phonon

scattering to dominate was therefore higher. Hence the shift of the current peak to higher temperatures as V_g was lowered. For $V_g < 20\text{ V}$ the currents diminished considerably, and the peak temperature shifted beyond 370 K as seen in Figure 7. For $V_g = 0\text{ V}$ no charge was electrostatically injected into the device and the $I_{ds}-T$ response seen in the inset to Figure 7 on a semi-log scale was like that reported by others^{25,26} in their undoped BBL thin film (i.e., I_{ds} increased monotonically with increasing T up to $T = 600\text{ K}$). The non-monotonic ($I_{ds}-T$) effects seen in Figure 7 were therefore a result of doping and were observed during the heating and cooling temperature runs of the $I_{ds}-V_g$ scans (at constant V_{ds}) as shown in Figure S4. Symmetric peak broadening with decreasing V_g as seen in Figure 7 reflected the weakening of phonon scattering on charge transport due to reduced channel current.

To confirm the “metallic-like” behavior seen during heating and cooling while recording $I_{ds}-V_g$ (Figure S4), we performed direct $I_{ds}-T$ experiments on the same device under identical bias conditions. Figure 8 shows the heating and cooling I_{ds} versus T traces in the temperature range $295\text{ K} < T < 370\text{ K}$, when the device was biased with $V_{ds} = 80\text{ V}$ and $V_g = 100\text{ V}$. The temperature and I_{ds} were first stabilized at 295 K , then the temperature was changed at a rate of 2 K/min during the heating and cooling run during the measurements. Under such conditions, the $I_{ds}-T$ trace upon heating showed the “metallic-like” response albeit with a broad plateau region, but the cooling trace was monotonic where the current decreased with decreasing temperature. In additional runs on the same device, the heating cycle always showed a non-monotonic $I_{ds}-T$ response, and the cooling run showed a monotonically decreasing response.

At the start of the heating run, with increasing temperature the trap concentration decreased leaving excess charge carriers in the channel. Charge transport due to thermal activation and phonon assisted hopping dominated and I_{ds} increased with increasing temperature. The activation energy, as calculated later in this temperature range, was consistent with that shown in the inset to Figure 6. As temperature increased, a relatively flat $I_{ds}-T$ response was observed where the effects of phonon scattering became comparable to that due to the other two processes. At still higher temperatures phonon scattering dominated and the current decreased with increasing temperature, consistent with Figure 7 and Figure S4. Upon cooling, the current continued to decrease, and the “metallic-like” response was not observed. Lowering T increased V_{th} and thus the trap concentration. The channel was also under electronic stress ($V_g = 100\text{ V}$) which is known to create metastable traps in the semiconductor.⁴⁵ Charge localization in these traps reduced

I_{ds} which further diminished the effects of phonon scattering. Hence the current decreased monotonically with decreasing temperature. Additional experiments like time-of-flight measurements may be needed to find the precise origin of this effect.

To compare activation energies we assumed that the temperature dependence for I_{ds} in Figure 8 was Arrhenius in nature, with μ replaced by I_{ds} in Equation (2). The inset to Figure 8 shows the data in an Arrhenius plot of I_{ds} versus inverse temperature, from which we calculated an activation energy of 156 meV during the start of the heating run and 316 meV during the cooling run. The lower value of 156 meV was close to that measured under identical conditions of V_{ds} and V_g as seen in the inset to Figure 6. The mechanisms governing charge transport were therefore consistent in both types of measurement that is, $I_{ds}-V_g$ or $I_{ds}-T$. The higher value of 316 meV also confirms the presence of deep traps within the polymer bandgap as discussed earlier. Our results show that dynamic filling of charge traps by a changing V_g at fixed temperature versus the filled traps condition with a constant V_g and changing T , could lead to different $I_{ds}-T$ response for heating and cooling runs. Charge transport in BBL was therefore dependent on the external electrical bias conditions, temperature cycling and was more complex as anticipated. More experiments are needed to better understand the precise origin of the non-monotonic high temperature “metallic-like” current response in BBL thin films. A possible application inspired by this work is to use BBL as a gate-controlled thermistor, whose temperature coefficient of the resistance and its switch in sign is controlled by an external bias voltage. Other applications of BBL include printed electronics⁴⁶ and solar based water oxidation⁴⁷ among others.

4 | CONCLUSIONS

Charge transport in BBL thin films were investigated using a FET configuration in the temperature range $150\text{ K} < T < 370\text{ K}$. Charge hopping, and charge activation were the dominant transport mechanisms for $T < 335\text{ K}$, while phonon back-scattering dominated at high temperatures. Assuming activated charge transport, the activation energies were found to lie in the range $100\text{ meV} < E_A < 400\text{ meV}$ indicating the existence of deep traps within the polymer bandgap of 1.8 eV. Two quasi-linear dependencies of the activation energy on V_g were also observed suggesting that multiple exponential-like energy distributions of trap states are present in BBL.

For $V_g > 20\text{ V}$, $I_{ds}(T)$ was non-monotonic, increasing with increasing temperature up to T_{max} above which it decreased. This decrease resembled “metallic-like” transport.

The temperature at which I_{ds} peaked shifted to higher values with lowering V_g due to reduced carrier charge concentration and deep trap filling. The mobility also showed a non-monotonic temperature response implying the presence of two different charge transport regimes in the above temperature range. While enhanced electron delocalization in BBL combined with increased carrier density could promote “metallic” behavior, what was observed was more likely localized charge transport limited by phonon scattering.

Direct measurements of I_{ds} with temperature for $V_{ds} = 80\text{ V}$ and $V_g = 100\text{ V}$ reproduced the “metallic-like” transport feature during the heating cycle but not during the cooling cycle. High electronic stress in BBL combined with increased trap density while cooling may be responsible for its absence. Our results revealed that charge transport in BBL was controlled by V_g and the energy distribution of traps. This could be due to enhanced carrier concentration combined with enhanced charge delocalization, the layered discrete structure, and a perpendicular layer attachment to the substrate. While the occurrence of “metallic-like” transport depended on V_g , its precise origin remains unclear and additional experiments are needed to better understand this effect.

AUTHOR CONTRIBUTIONS

Nicholas Pinto: Conceptualization (lead); formal analysis (equal); investigation (equal); supervision (lead); writing – original draft (equal). **Alejandro J. Cruz-Arzon:** Formal analysis (equal); investigation (equal); writing – original draft (equal). **William Serrano-Garcia:** Investigation (equal); writing – original draft (equal). **Nikita Gupta:** Investigation (supporting); supervision (supporting); writing – review and editing (supporting). **Alan T. Charlie Johnson:** Conceptualization (equal); writing – original draft (equal).

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Nicholas J. Pinto  <https://orcid.org/0000-0003-0954-6539>

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